

No	Qiyinlik darajasi	Savol	To'g'ri javob	Muqobil javob	Muqobil javob	Muqobil javob
1.	1	$Ax + By + C = 0$ ko'rinishdagi tenglama qanday ataladi?	*Tekislikda to'g'ri umumiy chiziqlarning tenglamasi.	To'g'ri chiziqlarning kanonik tenglamasi	To'g'ri chiziqlarning burchak koeffitsientli tenglamasi	To'g'ri chiziqlarning normal tenglamasi
2.	2	To'g'ri chiziqlarning umumiy tenglamasida x va y o'zgaruvchilar oldidagi koeffitsientlar qanday geometrik ma'noga ega.	* To'g'ri chiziqlar normal vektorining koordinatlari	To'g'ri chiziqlar yo'naltiruvchi vektorining koordinatlari	To'g'ri chiziqlarning Ox va Oy o'qlaridan ajratgan kesmalarining uzunliklari	Ixtiyoriy koordinatlar
3.	2	$Ax + By = 0, A \neq 0, B \neq 0$ tenglama bilan berilgan to'g'ri chiziqlar tekislikda koordinata sistemasiga nisbatan qanday joylashgan.	*to'g'ri chiziqlar koordinata boshidan o'tadi.	to'g'ri chiziqlar Oy o'qiga parallel.	to'g'ri chiziqlar Ox o'qiga parallel..	to'g'ri chiziqlar Ox o'qi bilan ustma-ust tushadi.
4.	2	$Ax + B = 0, B \neq 0$ tenglama bilan berilgan to'g'ri chiziqlar tekislikda koordinata sistemasiga nisbatan qanday joylashgan.	* to'g'ri chiziqlar Oy o'qiga parallel.	to'g'ri chiziqlar koordinata boshidan o'tadi.	to'g'ri chiziqlar Ox o'qi bilan ustma-ust tushadi.	to'g'ri chiziqlar Oy o'qi bilan ustma-ust tushadi.
5.	1	$\frac{x}{a} + \frac{y}{b} = 1$ tenglama to'g'ri chiziqlarning qanday ko'rinishdagi tenglamasi deyiladi ?	*To'g'ri chiziqlarning kesmalarga nisbatan tenglamasi	To'g'ri chiziqlarning umumiy tenglamasi	Kanonik ko'rinishdagi tenglamasi	Burchak koeffitsientli tenglamasi.
6.	1	$\frac{x-x_1}{m} = \frac{y-y_1}{n}$ tenglama to'g'ri chiziqlarning qanday ko'rinishdagi tenglamasi deyiladi, m, n sonlar qanday geometrik ma'noga ega.	*To'g'ri chiziqlarning kanonik tenglamasi, m va n yo'naltiruvchi vektorining koordinatalari	To'g'ri chiziqlarning kanonik tenglamasi, m va n normal vektorining koordinatalari.	To'g'ri chiziqlarning parametrik tenglamasi, m va n - haqiqiy sonlar.	To'g'ri chiziqlarning burchak koeffitsientli tenglamasi, m va n haqiqiy sonlar.
7.	1	$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1}$ ko'rinishdagi tenglama qanday ataladi?	*Berilgan ikki nuqtadan o'tuvchi to'g'ri chiziqlarning tenglamasi.	To'g'ri chiziqlarning kanonik tenglamasi.	To'g'ri chiziqlarning kesmalarga nisbatan tenglamasi.	To'g'ri chiziqlarning parametrik tenglamasi.
8.	2	$y = kx + b$ ko'rinishdagi tenglama to'g'ri chiziqlarning qanday ko'rinishdagi tenglamasi va k, b sonlar qanday geometrik ma'noga ega?	*To'g'ri chiziqlarning burchak koeffitsientli tenglamasi, k son to'g'ri chiziqlarning Ox o'qi bilan tashkil kilgan burchak	To'g'ri chiziqlarning burchak koeffitsientli tenglamasi, k son to'g'ri chiziqlarning Ox o'qi bilan tashkil kilgan burchak	To'g'ri chiziqlarning burchak koeffitsientli tenglamasi, $k = \cos \varphi$ b - ixtiyoriy son.	To'g'ri chiziqlarning parametrik tenglamasi, $k = \operatorname{ctg} \varphi$, b - to'g'ri chiziqlarning

			tangensiga teng, b - son to'g'ri chiziqning Oy o'qidan ajratgan kesmasi.	kotangensiga teng, b - son to'g'ri chiziqning Ox o'qidan ajratgan kesmasi.		Oy o'qidan ajratgan kesmasining uzunligi.
9.	2	$A_1x + B_1y + C_1 = 0$ va $A_2x + B_2y + C_2 = 0$ to'g'ri chiziqlar orasidagi burchakni qanday formula yordamida topiladi?	* $\cos \varphi = \frac{A_1A_2 + B_1B_2}{\sqrt{A_1^2 + B_1^2} \cdot \sqrt{A_2^2 + B_2^2}}$	$\sin \varphi = \frac{A_1A_2 + B_1B_2}{\sqrt{A_1^2 + B_1^2} \cdot \sqrt{A_2^2 + B_2^2}}$	$\cos \varphi = \frac{A_1B_2 + A_2B_1}{\sqrt{A_1^2 + A_2^2} \cdot \sqrt{B_1^2 + B_2^2}}$	$\operatorname{tg} \varphi = \frac{k_2 + k_1}{1 + k_1k_2}$
10.	2	$A_1x + B_1y + C_1 = 0$ va $A_2x + B_2y + C_2 = 0$ to'g'ri chiziqlarning parallellik va perpendikulyarlik shartlari mos ravishda quyidagicha:	* $\frac{A_1}{A_2} = \frac{B_1}{B_2}, A_1A_2 + B_1B_2 = 0$	$\frac{A_1}{B_2} = \frac{A_2}{B_1}, A_1A_2 + B_1B_2 = 0$	$\frac{A_1}{B_2} = \frac{A_2}{B_1}, A_1A_2 + B_1B_2 = 0$	$\frac{A_1}{A_2} = \frac{A_1}{B_1}, A_1A_2 + B_1B_2 = 0$
11.	2	$\frac{x-x_1}{m_1} = \frac{y-y_1}{n_1}$ va $\frac{x-x_2}{m_2} = \frac{y-y_2}{n_2}$ tenglamalar bilan berilgan to'g'ri chiziqlar orasidagi burchak qanday formula yordamida aniqlanadi?	* $\cos \varphi = \frac{m_1m_2 + n_1n_2}{\sqrt{m_1^2 + n_1^2} \cdot \sqrt{m_2^2 + n_2^2}}$	$\cos \varphi = \frac{m_1m_2 + n_1n_2}{\sqrt{m_1^2 + n_1^2} \cdot \sqrt{m_2^2 + n_2^2}}$	$\sin \varphi = \frac{m_1m_2 + n_1n_2}{\sqrt{m_1^2 + n_1^2} \cdot \sqrt{m_2^2 + n_2^2}}$	$\cos \varphi = \frac{m_1m_2 + n_1n_2}{\sqrt{m_1^2 + n_1^2} \cdot \sqrt{m_2^2 + n_2^2}}$
12.	2	$y = k_1x + b_1$ va $y = k_2x + b_2$ ko'rinishdagi tenglamalar bilan berilgan to'g'ri chiziqlar orasidagi burchak qanday formula yordamida aniqlanadi?	* $\operatorname{tg} \varphi = \frac{k_2 - k_1}{1 + k_1k_2}$	$\operatorname{tg} \varphi = \frac{k_2 - k_1}{1 - k_1k_2}$	$\operatorname{tg} \varphi = \frac{k_2 + k_1}{1 + k_1k_2}$	$\operatorname{tg} \varphi = \frac{1 - k_2k_1}{k_1 + k_2}$
13.	2	$y = k_1x + b_1$ va $y = k_2x + b_2$ tenglamalari bilan berilgan ikki to'g'ri chiziqning parallellik va perpendikulyarlik shartlarini ko'rsating.	* $k_1 = k_2, k_1k_2 = -1$	$k_1 + k_2 = 0, k_1k_2 = 1$	$k_1 = -k_2, \frac{k_1}{k_2} = 1$	$-k_1 = k_2, k_1 = \frac{1}{k_2}$
14.	1	$M_1(x_1, y_1)$ nuqtadan $Ax + By + C = 0$ to'g'ri chiziqgacha bo'lgan masofa qanday formula yordamida hisoblanadi?	* $d = \frac{ Ax_1 + By_1 + C }{\sqrt{A^2 + B^2}}$	$d = \frac{ Ax_1 + By_1 + C }{\sqrt{A^2 - B^2}}$	$d = \frac{\sqrt{A^2 - B^2}}{ Ax_1 + By_1 + C }$	$d = x_1 \cos \alpha + y_1 \cos \beta $
15.	2	$Ax + By + C = 0$ to'g'ri chiziq tenglamasi qanday shartda normal ko'rinishda bo'ladi?	* $A^2 + B^2 = 1, C \leq 0$	$A^2 + B^2 \neq 1, C \leq 0$	$A^2 - B^2 = 1$	$A + B = 1, C \leq 0$

16.	2	$A_1x + B_1y + C_1 = 0$ va $A_2x + B_2y + C_2 = 0$ to'g'ri chiziqlarning bir nuqtada kesishish shartini ko'rsating.	$* \begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix} \neq 0$	$\begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix} = 0$	$\begin{vmatrix} A_1 & C_1 \\ A_2 & C_2 \end{vmatrix} = 0$	$\begin{vmatrix} B_1 & C_1 \\ B_2 & C_2 \end{vmatrix} = 0$
17.	2	$Ax + By + Cz = 0$ tenglama bilan berilgan tekislik koordinata sistemasiga nisbatan qanday joylashgan?	*Koordinata boshidan o'tadi	Ox o'qiga parallel.	Oy o'qiga parallel	Oz o'qiga parallel.
18.	2	$By + Cz + D = 0$ tenglama bilan berilgan tekislik koordinata sistemasiga nisbatan qanday joylashgan?	*Ox o'qiga parallel.	Oz o'qiga parallel.	Oy o'qiga parallel.	Koordinata boshi orqali o'tadi.
19.	2	$Cz + D = 0$ tenglama bilan berilgan tekislik koordinata sistemasiga nisbatan qanday joylashgan?	*xOy koordinata tekisligiga parallel.	Ox o'qiga parallel.	Oy o'qiga parallel.	Koordinata boshidan o'tadi.
20.	2	Tenglamasi $By + D = 0$ bo'lgan tekislik koordinata tekisliklariga nisbatan qanday joylashgan?	*xOz koordinata tekisligiga parallel.	yOz koordinata tekisligiga parallel.	xOy koordinata tekisligiga parallel.	Oy o'qiga parallel.
21.	2	$Ax + D = 0$ tenglama bilan berilgan tekislik koordinata tekisliklariga nisbatan qanday joylashgan?	*yOz koordinata tekisligiga parallel.	Ox o'qiga parallel.	xOy koordinata tekisligiga parallel.	Oy o'qiga parallel.
22.	2	$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ ko'rinishdagi tenglama qanday ataladi va a, b, c sonlar qanday geometrik ma'noga ega?	* Tekislikning kesmalariga nisbatan tenglamasi, a, b, c -sonlar mos ravishda, Ox, Oy, Oz o'qlardan ajratgan kesmalarining uzunliklari.	Tekislikning umumiy tenglamasi; a, b, c - ixtiyoriy sonlar.	Tekislikning kesmalarga nisbatan tenglamasi, a, b, c - tekislik normal vektorining koordinatalari.	Tekislikning normal tenglamasi; a, b, c mos ravishda, Ox, Oy, Oz o'qlardan ajratgan kesmalarining uzunliklari.
23.	1	$A_1x + B_1y + C_1z + D_1 = 0$ va $A_2x + B_2y + C_2z + D_2 = 0$ tekisliklar orasidagi burchak quyidagi formula yordamida topiladi:	$* \cos \varphi = \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} \cdot \sqrt{A_2^2 + B_2^2 + C_2^2}}$	$\cos \varphi = \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} + \sqrt{A_2^2 + B_2^2 + C_2^2}}$	$\cos \varphi = \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} - \sqrt{A_2^2 + B_2^2 + C_2^2}}$	$\cos \varphi = \frac{A_1A_2 + B_1B_2 - C_1C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} \cdot \sqrt{A_2^2 + B_2^2 + C_2^2}}$

24.	2	$A_1x + B_1y + C_1z + D_1 = 0$ va $A_2x + B_2y + C_2z + D_2 = 0$ tenglamalar bilan berilgan tekisliklarning parallellik va perpendikulyarlik shartlarini ko'rsating.	$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2},$ *: $A_1A_2 + B_1B_2 + C_1C_2 = 0$	$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2},$ $A_1A_2 - B_1B_2 - C_1C_2 = 0$	$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}, A_1A_2 + B_1B_2 - C_1C_2 \neq 0$	$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}, A_1A_2 + B_1B_2 - C_1C_2 \neq 0$
25.	2	Bir to'g'ri chiziqda yotmaydigan uchta $M_1(x_1, y_1, z_1), M_2(x_2, y_2, z_2)$ va $M_3(x_3, y_3, z_3)$ nuqtalardan o'tuvchi to'g'ri chiziq tenglamasini yozing.	*: $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$	$\frac{x - x_1}{x_3 - x_2} = \frac{y - y_1}{y_3 - y_2} = \frac{z - z_1}{z_3 - z_2}$	$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$	$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ a & b & c \\ A & B & C \end{vmatrix} = 0$
26.	2	$M_1(x_1, y_1, z_1)$ nuqtadan $Ax + By + Cz + D = 0$ tekisligacha masofa qanday formula yordamida hisoblanadi.	*: $d = \frac{ Ax_1 + By_1 + Cz_1 + D }{\sqrt{A^2 + B^2 + C^2}}$	$d = \frac{Ax_1 + By_1 + Cz_1 + D}{\sqrt{A^2 + B^2 + C^2}}$	$d = \frac{ Ax_1 + By_1 + Cz_1 }{\sqrt{A^2 + B^2 + C^2}}$	$d = x_1 \cos \alpha + y_1 \cos \beta - z_1 \cos \gamma $
27.	1	$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$ ko'rinishdagi tenglama fazoda to'g'ri chiziqning qanday ko'rinishdagi tenglamasi deyiladi va l, m, n sonlarning geometrik ma'nosi qanday bo'ladi?	*Fazoda to'g'ri chiziqning kanonik tenglamasi l, m, n to'g'ri chiziq yo'naltiruvchi vektorining koordinatlari.	Fazoda to'g'ri chiziqning kanonik tenglamasi l, m, n to'g'ri chiziq normal vektorining koordinatlari.	To'g'ri chiziqning umumiy tenglamasi; l, m, n yo'naltiruvchi vektorining koordinatlari.	To'g'ri chiziqning geometrik tenglamasi; l, m, n - ixtiyoriy sonlar.
28.	1	Fazoda ikki $M_1(x_1, y_1, z_1)$ va $M_2(x_2, y_2, z_2)$ nuqtalardan o'tuvchi to'g'ri chiziq tenglamasini yozing.	*: $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$	$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$	$\frac{x - x_1}{y_2 - y_1} = \frac{y - y_1}{x_2 - x_1} = \frac{z - z_1}{1}$	$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$
29.	2	Berilgan ikkita $\frac{x - x_1}{l_1} = \frac{y - y_1}{m_1} = \frac{z - z_1}{n_1}$ va $\frac{x - x_2}{l_2} = \frac{y - y_2}{m_2} = \frac{z - z_2}{n_2}$ to'g'ri chiziqlar orasidagi burchak qanday formula yordamida topiladi?	*: $\cos \varphi = \frac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2} \cdot \sqrt{l_2^2 + m_2^2 + n_2^2}}$	$\sin \varphi = \frac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2} \cdot \sqrt{l_2^2 + m_2^2 + n_2^2}}$	$\cos \varphi = \frac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2} \cdot \sqrt{l_2^2 + m_2^2 + n_2^2}}$	$\cos \varphi = \frac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2} \cdot \sqrt{l_2^2 + m_2^2 + n_2^2}}$

30.	2	$\frac{x - x_1}{l_1} = \frac{y - y_1}{m_1} = \frac{z - z_1}{n_1}$ $\frac{x - x_2}{l_2} = \frac{y - y_2}{m_2} = \frac{z - z_2}{n_2}$ va ko‘rinishdagi tenglamalari bilan berilgan to‘g‘ri chiziqlarning parallellik va perpendikulyarlik shartlarini ko‘rsating.	$* \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2},$ $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$	$\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2},$ $l_1 l_2 + m_1 m_2 + n_1 n_2 \neq 0$	$\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2},$ $l_1 l_2 + m_1 m_2 + n_1 n_2 \neq 0$	$\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2},$ $l_1 l_2 - m_1 m_2 - n_1 n_2 \neq 0$
31.	2	$Ax + By + Cz + D = 0$ tekislik va $\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$ to‘g‘ri chiziq orasidagi burchak qanday formula yordamida hisoblanadi?	$* \sin \varphi = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \cdot \sqrt{l^2 + m^2 + n^2}}$	$\cos \varphi = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \cdot \sqrt{l^2 + m^2 + n^2}}$	$\sin \varphi = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \cdot \sqrt{l^2 + m^2 + n^2}}$	$\cos \varphi = \frac{Al + Bm + Cn}{\sqrt{A^2 + B^2 + C^2} \cdot \sqrt{l^2 + m^2 + n^2}}$
32.	2	$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$ to‘g‘ri chiziq va $Ax + By + Cz + D = 0$ tekislik umumiy nuqtaga ega bo‘lmaslik (parallellik) shartini ko‘rsating.	$* Al + Bm + Cn = 0, Ax_0 + By_0 + Cz_0 + D \neq 0$	$Al + Bm + Cn = 0, Ax_0 + By_0 + Cz_0 + D = 0$	$\frac{Ax_0}{l} + \frac{By_0}{m} + \frac{Cz_0}{n} + \frac{D}{1} = 0$	$\frac{A}{l} = \frac{B}{m} = \frac{C}{n}$
33.	2	$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$ to‘g‘ri chiziqning $Ax + By + Cz + D = 0$ tekislikka perpendikulyarlik shartini ko‘rsating.	$* \frac{A}{l} = \frac{B}{m} = \frac{C}{n}$	$Al + Bm + Cn = 0, Ax_0 + By_0 + Cz_0 + D \neq 0$	$\frac{Ax_0}{l} + \frac{By_0}{m} + \frac{Cz_0}{n} + \frac{D}{1} = 0$	$\frac{A}{l} = \frac{B}{m} = \frac{C}{n}$
34.	3	$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$ to‘g‘ri chiziqning $Ax + By + Cz + D = 0$ tekislikda yotish shartini ko‘rsating.	$* Al + Bm + Cn = 0,$ $Ax_0 + By_0 + Cz_0 + D = 0$	$Al + Bm + Cn = 0, Ax_0 + By_0 + Cz_0 + D \neq 0$	$\frac{Ax_0}{l} + \frac{By_0}{m} + \frac{Cz_0}{n} + \frac{D}{1} = 0$	$\frac{A}{l} = \frac{B}{m} = \frac{C}{n}$
35.	3	$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$ to‘g‘ri	$* Al + Bm + Cn \neq 0$	$Al + Bm + Cn = 0, Ax_0 + By_0 + Cz_0 + D \neq 0$	$\frac{Ax_0}{l} + \frac{By_0}{m} + \frac{Cz_0}{n} + \frac{D}{1} = 0$	$Al + Bm + Cn = 0$

		chiziqning $Ax + By + Cz + D = 0$ tekislik bilan bir nuqtada kesishish shartini ko'rsating.				
36.	3	$A_1x + B_1y + C_1z + D = 0, A_2x + B_2y + C_2z + D_2 = 0$ va $A_3x + B_3y + C_3z + D_3 = 0$ tenglamalar bilan berilgan tekisliklarning bir nuqtada kesishish shartini ko'rsating.	$\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = 0$	$\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} \neq 0$	$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$	$\frac{A_1}{A_3} = \frac{B_1}{B_3} = \frac{C_1}{C_3}$
37.	1	$\vec{a}$ va $\vec{b}$ vektorlarning skalyar ko'paytmasi quyidagicha bo'ladi ...	$* \vec{a} \vec{b} = \vec{a} \cdot \vec{b} \cdot \cos \varphi;$	$\vec{a} \vec{b} = \vec{a} \cdot \vec{b} \cdot \sin \varphi$	$\vec{a} \vec{b} = \vec{a} \cdot \vec{b} \cdot \operatorname{tg} \varphi$	$\vec{a} \vec{b} = \vec{a} \cdot \vec{b} \cdot \operatorname{ctg} \varphi$
38.	1	$\vec{a}$ va $\vec{b}$ vektorlar orasidagi burchak quyidagi formula yordamida aniqlanadi:	$* \cos \varphi = \frac{\vec{a} \vec{b}}{ \vec{a} \cdot \vec{b} }$	$\sin \varphi = \frac{ \vec{a} \vec{b} }{\sqrt{a^2} \sqrt{b^2}}$	$\operatorname{ctg} \varphi = \frac{\vec{a} \vec{b}}{\sqrt{a^2} \sqrt{b^2}}$	$\operatorname{tg} \varphi = \frac{\vec{a} \vec{b}}{ \vec{a} \vec{b} }$
39.	2	Koordinatalari bilan berilgan $\vec{a}$ va $\vec{b}$ vektorlarning vektor ko'paytmasi quyidagicha bo'ladi:	$* [\vec{a}, \vec{b}] = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$	$[\vec{a}, \vec{b}] = \vec{a} \vec{b} \cos \varphi$	$[\vec{a}, \vec{b}] = \sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2} \sin \varphi$	$[\vec{a}, \vec{b}] = \vec{a} \vec{b} \operatorname{ctg} \varphi$
40.	1	$\vec{a}$ va $\vec{b}$ vektorlarning vektor ko'paytmasi bo'lgan vektorning uzunligi quyidagiga teng.	$* [\vec{a}, \vec{b}] = \vec{a} \vec{b} \sin \varphi$	$ [\vec{a}, \vec{b}] = \vec{a} \vec{b} \operatorname{tg} \varphi$	$ [\vec{a}, \vec{b}] = \vec{a} \vec{b} \cos \varphi$	$ [\vec{a}, \vec{b}] = \vec{a} \vec{b} \operatorname{ctg} \varphi$
41.	2	$\vec{a}$ va $\vec{b}$ vektorlarning ortogonallik shartini ko'rsating.	$* \vec{a} \cdot \vec{b} = 0$	$[\vec{a} \cdot \vec{b}] = 0$	$\vec{a} + \vec{b} = 0$	$\vec{a} \cdot \vec{b} = 0$
42.	2	$\vec{a}, \vec{b}, \vec{c}$ vektorlarning komplanarlik shartini ko'rsating.	$* \vec{a} \vec{b} \vec{c} = 0$	$(\vec{a} - \vec{b}) \cdot \vec{c} = 0$	$(\vec{a} + \vec{b}) \cdot \vec{c} = 0$	$(\vec{a} \cdot \vec{b}) \vec{c} = 0$
43.	1	$\vec{a} = 4\vec{i} + 7\vec{j} + 3\vec{k}$ va $\vec{b} = 3\vec{i} - 5\vec{j} + \vec{k}$ vektorlarning skalyar ko'paytmasi quyidagiga teng.	$* \vec{a} \vec{b} = -20;$	$\vec{a} \vec{b} = 20;$	$\vec{a} \vec{b} = -50;$	$\vec{a} \vec{b} = 30$
44.	1	$\vec{a} = \vec{i}$ va $\vec{b} = \vec{i} + \vec{j}$ vektorlar orasidagi burchak quyidagiga teng.	$* \varphi = 45^\circ;$	$\varphi = 90^\circ;$	$\varphi = 30^\circ;$	$\varphi = 0^\circ$
45.	2	Koordinata boshidan $x + y - \sqrt{2} = 0$ to'g'ri chiziqqacha bo'lgan masofa quyidagiga	$* d = 1;$	$d = 2;$	$d = \sqrt{2};$	$d = 8$

		teng:				
46.	2	$M(1,2)$ nuqtadan $2x - y - \sqrt{5} = 0$ to'g'ri chiziqqacha bo'lgan masofa quyidagilardan biriga teng.	$*d = 1$	$d = \sqrt{5}$	$d = 0$	$d = \frac{1}{\sqrt{5}}$
47.	2	$3x + 4y + 1 = 0$ va $4x - 3y - 5 = 0$ to'g'ri chiziqlar orasidagi burchak quyidagilardan biriga teng.	$*\varphi = 90^0$;	$\varphi = 60^0$;	$\varphi = 30^0$;	$\varphi = 0^0$
48.	1	$y = 3x - 5$ to'g'ri chiziqning absissasi $x_0 = -4$ ga teng bo'lgan nuqtaning ordinatasi y_0 ni toping.	$*-17$	-4	17	7
49.	1	$C(3;2)$ nuqtaga koordinata boshiga nisbatan simmetrik bo'lgan nuqtani toping.	$*(-3;-2);$	$(3;2);$	$(3;-2);$	$(2;3)$
50.	2	$M\left(6; \frac{\pi}{2}\right)$ nuqta qutb koordinatalarida berilgan, uning dekart kordinatalarini toping.	$*(0;6)$	$(3;-2)$	$(-2;4)$	$(6;0)$
51.	2	Uchlari $A(0;0)$, $B(4;0)$, $C(0;6)$ nuqtalarda bo'lgan uchburchakning yuzini toping..	$*12$	14	13	6
52.	2	Uchlari $A(-2;0)$, $B(0;7)$, $C(0;0)$ nuqtalarda bo'lgan uchburchakning yuzini toping.	$*7$	3	4	14
53.	3	Absissa o'qida $A(-3;4;-8)$ nuqtadan 12 birlik uzoqlikda bo'lgan nuqtani toping.	$*(5;0;0)$ va $(-11;0;0)$	$(-3;0;0)$ va $(-8;1;0)$	$(5;0;0)$ va $(-5;0;0)$	To'g'ri javob yo'q
54.	1	$A(1;-3)$, $B(3;-5)$ nuqtalar $\overline{AB}$ kesmaning oxirlari bo'lsa, kesma o'rtasining koordinatalarini toping.	$*(2;-4)$	$(0;2)$	$(5;8)$	$(-2;2)$
55.	1	Agar $\vec{a} = \{1;3;-1\}$, $\vec{b} = \{2;1;4\}$ bo'lsa $\vec{c} = \vec{a} + \vec{b}$ ni toping.	$*\vec{c} = \{3;4;3\}$;	$\vec{c} = \{0;2;1\}$;	$\vec{c} = \{5;0;3\}$;	$\vec{c} = \{2;3;-1\}$

56.	1	$A = (3;1;5), B = (1;2;2)$ bo'lsa $\overline{AB}$ vektorning koordinatalarini toping.	$*\{-2;1;-3\}$;	$\{1;2;3\}$;	$\{0;1;4\}$;	$\{-1;2;3\}$
57.	2	$\vec{a} = \{2;-1;3\}, \vec{b} = \{-6;3;-9\}$ vektorlar qanday o'zaro munosabatda bo'ladi?	$*\vec{a}$ va $\vec{b}$ kollinear bo'ladi.	$\vec{a} \perp \vec{b}$	$\vec{a} = \vec{b}$	$\vec{a}$ va $\vec{b}$ kollinear emas.
58.	1	$\vec{a} = \{x_1 y_1 z_1\}$ va $\vec{b} = \{x_2; y_2; z_2\}$ vektorlarning skalyar ko'paytmasi qanday formula yordamida topiladi?	$*\vec{a}\vec{b} = x_1x_2 + y_1y_2 + z_1z_2$	$\vec{a}\vec{b} = x_1^2x_2^2 + y_1^2y_2^2 + z_1^2z_2^2$	$\vec{a}\vec{b} = (x_1 + y_1 + z_1)(x_2 + y_2 + z_2)$	$\vec{a}\vec{b} = (x_1 + y_1 + z_1)(x_2 + y_2 + z_2)$
59.	2	Agar $xy > 0$ bo'lsa $M(x, y)$ nuqta qaysi chorakda joylashgan?	$*I$ va III	II va III	III va IV	I va IV
60.	2	Agar $xy < 0$ bo'lsa $M(x, y)$ nuqta qaysi chorakda joylashgan?	$*II$ va IV	I va III	III va IV	I va IV
61.	3	$\vec{a} = \{3;0;-4\}$ va $\vec{b} = \{1;-2;2\}$ vektorlar orasidagi burchak sinusini toping.	$*\frac{2\sqrt{2}}{3}$;	$\frac{2\sqrt{3}}{3}$;	$\frac{2}{3}$;	$\frac{3}{4}$;
62.	3	$\vec{a}\vec{b} = 42$ bo'lgan holda, $\vec{a} = \{4;2;-1\}$, vektorga kollinear $\vec{b}$ vektorni toping.	$*\vec{b} = \{8;4;-2\}$	$\vec{b} = \{2;1;-1\}$	$\vec{b} = \{-4;-2;1\}$	$\vec{b} = \{2;4;-1\}$
63.	2	$\vec{a} = \{-2;-1;1\}, \vec{b} = \{4;-4;1\}, \vec{c} = \{4;-6;2\}$ vektorlarning aralash ko'paytmasini toping	$*0$;	6 ;	12 ;	4 ;
64.	2	$\vec{a} = \{-1;3;4\}, \vec{b} = \{2;5;2\}, \vec{c} = \{1;2;3\}$ vektorlarga yasalgan parallelepipedning hajmini toping.	$*27$;	-27 ;	54 ;	$13,5$
65.	2	$\vec{a} = \{2;-3;-1\}$ vektor oxirining koordinatalari $(1;-1;2)$ nuqtada bo'lsa, boshining koordinatalarini toping.	$*(-1;2;3)$;	$(-1;3;2)$;	$(0;-1;2)$;	$(3;2;-1)$

66.	1	$M(0; -4)$ nuqtaning qutb koordinatasini toping.	$* \left(4; \frac{3\pi}{2}\right);$	$\left(4; \frac{\pi}{4}\right);$	$\left(4; \frac{\pi}{2}\right);$	$(4; 45^0);$
67.	3	Ekstsentrisiteti $e = 2$ bo'lgan giperbolaning asimptotalari orasidagi burchakni toping.	$* \alpha = 120^0$	$\alpha = 90^0$	$\alpha = 45^0$	$\alpha = 30^0$
68.	1	$\frac{x^2}{225} - \frac{y^2}{64} = 1$ giperbolaning fokuslarini toping.	$F_1(-17, 0), F_2(17, 0)$	$F_1(-17, 0), F_2(0, -17)$	$F_1(0, -17), F_2(0, 17)$	$F_1(-15, 0), F_2(15, 0)$
69.	1	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ giperbolaning asimptotalari tenglamalarini ro'rsating.	$* y = \pm \frac{b}{a} x$	$y = \pm \frac{c}{a} x$	$y = \pm \frac{be}{a} x$	$y = \pm \frac{a}{e} x$
70.	1	$y^2 = 2px$ parabolaga $M_0(x_0, y_0)$ nuqtasidagi urinmasi:	$* yy_0 = p(x + x_0)$	$y - y_0 = p(x - x_0)$	$y + y_0 = p(x + x_0)$	$yy_0 = p(x - x_0)$
71.	1	Quyidagi nuqtalardan qaysi biri $y^2 = 8x$ parabolaga tegishli?	$* A(2, -4)$	$B(4, 2)$	$M(-2, -4)$	$P(-2, 4)$
72.	2	$x^2 + y^2 - 6x = 0$ aylananing markazi S radiusi r ni toping.	$* S(3, 0), r = 3$	$S(0, 3), r = 4$	$S(-3, 0), r = 2$	$S(3, 0), r = 2$
73.	1	Quyidagi nuqtalardan qaysi biri $x^2 + y^2 - 1 = 0$ aylanaga tegishli?	$* S(-1, 0)$	$T(3, 0)$	$R(3, 1)$	$V(-2, 0)$
74.	2	$x^2 + y^2 = 1$ aylanaga $M_0(1, 0)$ nuqtada o'tkazilgan urinma tenglamasi:	$* x = 1$	$x = -1$	$y = 1$	$y = -1$
75.	3	Radiusi $r = 3$ markazi $S(3, 0)$ nuqtada bo'lgan aylana tenglamasi:	$* x^2 + y^2 - 6x = 0$	$x^2 + y^2 - 8x = 0$	$x^2 + y^2 - 1 = 0$	$x^2 + y^2 - 4x = 0$
76.	2	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ellipsga $M_0(x_0, y_0)$ nuqtasida o'tkazilgan urinma tenglamasini toping.	$* \frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$	$\frac{x - x_0}{a^2} + \frac{y - y_0}{b^2} = 1$	$\frac{x - x_0}{a^2} + \frac{y - y_0}{b^2} = 0$	$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 0$
77.	1	Quyidagi nuqtalardan qaysi biri $\frac{x^2}{9} + \frac{y^2}{25} = 1$ ellipsga tegishli?	$* K(\sqrt{3}, 5\sqrt{\frac{2}{3}})$	$A(6, 1)$	$B(-1, 7)$	$L(\sqrt{\frac{17}{2}}, 5)$
78.	2	Ikkinchi tartibli chiziqli markazini toping $5x^2 + 8xy + 5y^2 - 18x - 18y + 11 = 0$.	$* O'(1, 1)$	$O'(-1, -1)$	$O'(2, 2)$	$O'(1, -1)$

79.	2	Quyidagi tenglamalardan qaysi biri ellipsoidni aniqlaydi?	$* \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
80.	2	Quyidagi tenglamalardan qaysi biri bir pallali giperboloidni aniqlaydi?	$* \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
81.	2	Quyidagi tenglamalardan qaysi biri ikki pallali giperboloidni aniqlaydi?	$* \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
82.	2	Quyidagi tenglamalardan qaysi biri konusni aniqlaydi?	$* \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
83.	2	Quyidagi tenglamalardan qaysi biri giperbolik paraboloidni aniqlaydi?	$* \frac{x^2}{a^2} - \frac{y^2}{b^2} = 2pz$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
84.	2	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2pz$ tenglama bilan berilgan sirt qanday nomlanadi ?	* elliptik paraboloid	Elliptik silindr	Giperbolik silindr	Parabolik silindr
85.	2	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ tenglama bilan berilgan sirt qanday nomlanadi ?	*Elliptik silindr	elliptik paraboloid	Giperbolik silindr	Parabolik silindr
86.	2	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ tenglama bilan berilgan sirt qanday nomlanadi ?	*Giperbolik silindr	elliptik paraboloid	Elliptik silindr	Parabolik silindr
87.	2	$y^2 = 2px$ tenglama bilan berilgan sirt qanday nomlanadi ?	*Parabolik silindr	elliptik paraboloid	Elliptik silindr	Giperbolik silindr
88.	3	Uchi A(1,1) nuqtada, parametri p=2 ga teng bo'lgan va B(2,3) nuqtadan o'tuvchi parabola tenglamasini tuzing.	$* y^2 - 4x - 2y + 5 = 0$	$y^2 - 2x - 2y + 3 = 0$	$y^2 + 2x^2 - 2y + 3 = 0$	$y^2 - 2x^2 - 2y + 3 = 0$
89.	3	Markazi koordinata boshida, uchlari A(3,0) va B(0,-2) nuqtalarda bo'lgan ellipsning tenglamasini toping.	$* \frac{x^2}{9} + \frac{y^2}{4} = 1$	$\frac{x^2}{9} + \frac{y^2}{2} = 1$	$\frac{x^2}{3} + \frac{y^2}{4} = 1$	$\frac{x^2}{7} - \frac{y^2}{5} = 1$
90.	3	$\frac{x^2}{9} + \frac{y^2}{4} = 1$ tenglama bilan berilgan ellipsning A(3,0) nuqtasidan o'tuvchi urinmasining tenglamasini tuzing.	$* x - 3 = 0$	$x - 4 = 0$	$x + 4 = 0$	$x = 0$

91.	2	Berilgan girepbolaning asimptotalarini toping: $\frac{x^2}{12} - \frac{y^2}{6} = 1$.	$* y = \frac{x}{2}, y = -\frac{x}{2}$	$y = \frac{x}{5}, y = -\frac{x}{2}$	$y = \frac{x}{3}, y = \frac{x}{4}$	$y = \frac{x}{3}$
92.	2	$M(3, -2, 3)$ va $N(-2, 0, 4)$ nuqtalar berilgan bo'lsa, MN kesmani 1:2 nisbatda bo'luvchi nuqtaning koordinatalarini toping.	$* (\frac{4}{3}, -\frac{4}{3}, \frac{10}{3})$	$(3, 0, 3)$	$(3, -2, 0)$	$(3, -\frac{2}{3}, 3)$
93.	3	$\vec{a} = 5\vec{p} + 2\vec{q}$ va $\vec{b} = \vec{p} - 3\vec{q}$ vektorlarga qurilgan parallelogrammning katta diagonalli uzunliklarini toping, bu yerda $ \vec{p} = 2\sqrt{2}$, $ \vec{q} = 3$ va $(\vec{p}, \vec{q}) = \pi/2$.	$* \vec{a} + \vec{b} = 9\sqrt{11}$	$ \vec{a} + \vec{b} = \sqrt{30}$	$ \vec{a} + \vec{b} = \sqrt{40}$	$ \vec{a} + \vec{b} = \sqrt{17}$
94.	3	Agar $\vec{p}$ va $\vec{q}$ o'zaro perpendikulyar birlik vektorlar bo'lsa, $\vec{a} = 3\vec{p} + 2\vec{q}$ va $\vec{b} = \vec{p} + 5\vec{q}$ vektorlar orasidagi burchak kosinusini toping	$* \frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{3}$	$\frac{\sqrt{2}}{5}$	$\frac{\sqrt{3}}{2}$
95.	3	$\vec{m}$ va $\vec{n}$ - o'zaro perpendikulyar birlik vektorlar bo'lsa, $\vec{a} = 3\vec{m} - 4\vec{n}$ vektorning uzunligini toping.	$* 5$	4	6	3
96.	3	$\vec{a} = 3\vec{m} + 5\vec{n}$, $\vec{b} = \vec{m} - 2\vec{n}$ va $\vec{c} = 2\vec{m} + 7\vec{n}$ vektorlarga qurilgan parallelepipedning hajmini hisoblang, bu yerda $ \vec{m} = 1/2$, $ \vec{n} = 3$, $(\vec{m}, \vec{n}) = 135^\circ$	$* 0$	1	3	5
97.	3	$\vec{A} = 3\vec{P} + 2\vec{Q} - 5\vec{R}$, $\vec{B} = \vec{P} - \vec{Q} + 4\vec{R}$ va $\vec{C} = \vec{P} - 3\vec{Q} + \vec{R}$ vektorlarga qurilgan parallelepipedning balandligini hisoblang, bunda asos sifatida $\vec{A}$ va $\vec{B}$ vektorlarga qurilgan parallelogramm olingan. $\vec{P}, \vec{Q}, \vec{R}$ - o'zaro perpendikulyar birlik vektorlar.	$* \frac{44}{\sqrt{323}}$	$\frac{\sqrt{2}}{5}$	$\frac{\sqrt{21}}{5}$	$\sqrt{\frac{12}{5}}$
98.	3	$\vec{AB} = \vec{m} + 2\vec{n}$ va $\vec{AD} = \vec{m} - 3\vec{n}$ vektorlarga qurilgan parallelogrammning	$* \frac{75}{2}$	$\frac{25}{2}$	$\frac{105}{2}$	$\frac{75}{4}$

		yuzasini hisoblang, bu yerda $m = 5$, $n = 3$ va $(\hat{m}^p, \hat{n}^p) = \pi / 6$.				
99.	2	Sferani markazidan o'tuvchi tekislik bilan kesganda ... hosil bo'ladi.	*Aylana	Ellips	Nuqta	To'g'ri chiziq
100.	2	Sferani markazidan o'tuvchi tekislik bilan kesganda hosil bo'lgan aylananing radiusi ...	*berilgan sfera radiusiga teng	berilgan sfera radiusidan ikkiki barobar kata	berilgan sfera radiusining yarmiga teng	aniqlab bo'lmaydi
101.	1	Yevklidning "Negizlar" asari nechta kitobdan iborat bo'lgan.	*13 ta	15 ta	10 ta	8 ta
102.	1	Gilbert aksiomalar sistemasi nechta guruhga bo'linadi.	*5 ta	4 ta	3 ta	6 ta
103.	2	Gilbert aksiomalar sistemasida Pash aksiomasi qaysi guruh aksiomalariga kiradi.	*Tartib aksiomalari	Parallellik aksiomasi	5 guruh aksiomasi	Tenglik aksiomalari
104.	2	II guruh aksiomalar turkumiga kiruvchi Pash aksiomasini keltirilgan javobni ko'rsating.	*Agar uchburchakning birorta ham uchidan o'tmaydigan va uning tekisligida yotuvchi to'g'ri chiziq uning bir tomoni bilan umumiy nuqtaga ega bo'lsa, u holda bu to'g'ri chiziq uchburchakning boshqa biror tomoni bilan ham umumiy nuqtaga ega	Parallel to'g'ri chiziqlar kesishmaydi	Bir tekislikda yotuvchi ikki nuqtadan yagona to'g'ri chiziq otadi	To'g'ri javob yo'q.
105.	2	Teng yonli uchburchakning asosidagi burchaklar tengdir. Ushbu jumla qaysi guruh aksiomalariga teng kuchli.	*Kongurentlik (Tenglik) guruh aksiomalariga.	Parallellik guruhi	Tegishlilik guruhi	Tartib guruhi
106.	2	Vertikal burchaklar o'zaro teng. Ushbu jumla qaysi guruh aksiomalariga teng kuchli.	*Kongurentlik (Tenglik) guruh aksiomalariga.	Parallellik guruhi	Tegishlilik guruhi	Tartib guruhi
107.	2	Gilbert aksiomalar sistemasida Dedikend aksiomasi qaysi guruhga tegishli.	*Uzluksizlik aksiomalari	Parallellik aksiomalari	1 ta	To'g'ri javob yo'q
108.	2	Gilbert aksiomalar sistemasida Kantor teoremasi qaysi guruhga tegishli.	*Uzluksizlik aksiomalari	Parallellik aksiomalari	5 ta	To'g'ri javob yo'q
109.	2	Gilbert aksiomalar sistemasida Kantor teoremasi qaysi guruhga tegishli.	*Uzluksizlik aksiomalari	Parallellik aksiomalari	2 ta	To'g'ri javob yo'q

110.	1	Parallellik aksiomasi to'g'ri ko'rsatilgan javobni belgilang.	*To'g'ri chiziq tashqarisidagi nuqtadan o'tib, bu to'g'ri chiziq bilan kesishmaydigan to'g'ri chiziq bittadan ortiq emas.	Tegishlilik guruhi	Agar uchburchakning birorta ham uchidan o'tmaydigan va uning tekisligida yotuvchi to'g'ri chiziq uning bir tomoni bilan umumiy nuqtaga ega bo'lsa, u holda bu to'g'ri chiziq uchburchakning boshqa biror tomoni bilan ham umumiy nuqtaga ega	Bir tekislikda yotuvchi ikki nuqtadan yagona to'g'ri chiziq otadi
111.	2	Gilbert aksiomalar sistemasida quyidagi teorema qaysi guruh aksiomalariga tegishli. Uchburchakning ichki burchaklari yig'indisi 180^0 ga teng.	*Parallellik aksiomalari.	Tegishlilik aksiomalari.	1 ta.	Tartib aksiomalari.
112.	2	Aylanaga ichki chizilgan olti burchakning tomoni ... ga teng.	*Shu aylananing radiusiga	Shu aylananing diametriga	Shu aylananing vatariga	6
113.	2	Lobachevskiy tekisligida uchburchak ichki burchaklari yig'indisi qanday?	* 180^0 dan kichik	180^0 ga teng	200^0 ga teng	300^0 ga teng
114.	2	... tekisligida har qanday to'rtburchak ichki burchaklarining yig'indisi 360^0 dan kichik bo'lib, bu son har xil to'rtburchaklar uchun har xildir.	*Lobachevskiy	Yevklid	To'g'ri javob yo'q	7
115.	1	Lobachevskiy tekisligidagi har qanday ... burchak parallellik burchagidir.	*O'tkir burchak	180^0	190^0	200^0
116.	2	Lobachevskiy tekisligida bitta to'g'ri chiziqqa perpendikulyar to'g'ri chiziqlar ... deyiladi.	*Uzoqlashuvchi to'g'ri chiziqlar	Kesishuvchi to'g'ri chiziqlar	Nuqta	Kesma
117.	2	Ushbu $\Pi(p) = 2arctg e^{\frac{p}{k}}$ funksiyaga ... funksiyasi deyiladi.	*Lobachevskiy	Kvadrat	Kubik	Chiziqli
118.	2	Aylana, ekvidistant chiziqlar, oritsikl chiziqlar qaysi geometriyaga tegishli.	*Lobachevskiy geometriyasi	Yevklid geometriyasi	Bunday chiziqlar mavjud emas	To'g'ri javob yo'q
119.	2	Beltrami-Kleyn va Puankare modellari qaysi geometriya interpretatsiyasida qo'llaniladi.	*Lobachevskiy geometriyasi	Yevklid geometriyasi	Bunday modellar mavjud emas	To'g'ri javob yo'q

120.	1	Chizg'ich, sirkul, chizmachilik uchburchagi kabi yasash quollari yordamida yasashni talab etgan masala ...deyiladi.	*Konstruktiv masala	Chiziqli tenglama	Bunday masala yo'q	To'g'ri javob yo'q
121.	1	Quyidagi yasash masalasi yasashga doir qanday masala hisoblanadi. Berilgan burchakka teng burchak yasash.	*Elementar	Murakkab	Yasashga doir bunday masala yo'q	To'g'ri javob yo'q
122.	2	Quyidagi yasash masalasi yasashga doir qanday masala hisoblanadi. Berilgan nuqtadan berilgan aylanaga urinma o'tkazish.	*Elementar	Murakkab	Yasashga doir bunday masala yo'q	To'g'ri javob yo'q
123.	2	Quyidagi yasash masalasi yasashga doir qanday masala hisoblanadi. Berilgan kesmani berilgan nisbatda bo'lish.	*Elementar	Murakkab	Yasashga doir bunday masala yo'q	To'g'ri javob yo'q
124.	2	Teng yonli to'g'ri burchakli uchburchakning o'tkir burchagi bissektrisasi qarshisidagi katetni to'g'ri burchak uchidan hisoblaganda qanday nisbatda bo'ladi?	$*\sqrt{2}:2$	1:2	2:1	3:2
125.	2	ABC uchburchakning AB, BC va CA tomonlarida olingan M, N va P nuqtalar shu tomonlarni 1:2 nisbatda bo'ladi. Agar ABC uchburchakning yuzi S ga teng bo'lsa, MNP uchburchakning yuzini toping.	$*\frac{1}{3}S$	$\frac{1}{2}S$	$\frac{1}{4}S$	$\frac{1}{6}S$
126.	1	To'g'ri burchakli uchburchak katetlarining gipotenuzadagi proyeksiyalari 8 va 2 ga teng. Uchburchakning yuzini toping.	*20	30	40	50
127.	1	Perimetrlari 24 va 36 bo'lgan ikki o'xshash uchburchakdan birining yuzi ikkinchisidan 10 ga ortiq. Kichik uchburchakning yuzini toping	*8	10	12	16
128.	2	Yuzi 48 ga teng bo'lgan ABC uchburchakning AC tomoni D nuqtada $AD:DC = 1:7$ kabi nisbatda bo'linadi. ABD uchburchakning yuzini aniqlang	*6	8	10	12
129.	1	Kvadratning tomonini necha marta	*2	4	6	8

		kamaytirganda yuzi 4 marta kichrayadi?				
130.	2	To'g'ri to'rtburchakning perimetri 60 ga teng, bir tomoni boshqa tomonidan 6 ga ko'p. To'g'ri to'rtburchakning yuzini toping.	*216	108	144	180
131.		Rombning uchidan tushirilgan balandligi uning tomonini, o'tkir burchagi uchidan boshlab hisoblaganda, 3 va 2 ga teng kesmalarga bo'ladi. Rombning yuzini toping.	*20	30	40	60
132.	2	Parallelogrammning o'tkir burchagi 60^0 ga teng. Uning kichik diagonal katta tomoni bilan 30^0 li burchak tashkil qiladi. Parallelogrammning katta tomoni 20 ga teng. Uning yuzini toping.	$*100\sqrt{3}$	$10\sqrt{3}$	$200\sqrt{3}$	$300\sqrt{3}$
133.	2	Teng yonli trapetsiyaning kichik asosi 3 ga, perimetri 42 ga teng. Uning diagonal o'tmas burchagini teng ikkiga bo'ladi. Trapetsiyaning o'rta chizig'ini toping.	*8	16	4	24
134.	1	Qavariq ko'pburchak ichki burchaklarining va bitta tashqi burchagining yig'indisi $\frac{23\pi}{2}$ ga teng. Ko'pburchakning nechta tomoni bor?	*13	14	12	15
135.	2	Radiusi 5 ga teng bo'lgan aylana yoyining uzunligi radiusi 2 ga teng aylana uzunligiga teng bo'lsa, hosil bo'lgan markaziy burchakni toping.	* 144^0	120^0	135^0	150^0
136.	2	Barcha nuqtalarining koordinatalari $x^2 + y^2 \leq 4x + 6y$ tengsizlikni qanoatlantiradigan tekis figuraning yuzini toping	* 13π	12π	11π	10π
137.	2	Tog'ri burchakli uchburchak katetlarining gepotenuzasidagi proeksiyalari 9 va 16 ga	*5	4	3	6

		teng. Uchburchakka ichki chizilgan aylananing radiusi qancha?				
138.	3	Ikkita burchagi 45^0 va 75^0 bo'lgan uchburchakning uzunligi bo'yicha o'rtacha tomoni $3\sqrt{3}$ ga teng. Uchburchakka tashqi chizilgan aylananing radiusini toping	*3	6	9	4
139.	2	Diagonallari 12 va 16 ga teng bo'lgan rombga ichki chizilgan aylananing radiusini toping.	*4,8	2,4	9,6	6
140.	2	O'tkir burchagi 30^0 bo'lgan to'g'ri burchakli trapetsiya diametri 8 ga teng aylanaga tashqi chizilgan. Trapetsiyaning yuzini toping.	*96	106	98	94
141.	2	Bir nuqtadan tekislikka uzunliklari 4 va 8 bo'lgan ikkita og'ma tushirilgan. Og'malar proyeksiyalarining nisbati 1:7 ga teng. Berilgan nuqtadan tekislikkacha bo'lgan masofani toping.	$*\sqrt{15}$	15	4	9
142.	2	Uchburchakli to'g'ri prizma asosining tomonlari 15; 20 va 25 ga, yon qirradi asosining balandligiga teng. Prizmaning hajmini toping	*1800	900	1200	1440
143.	2	Uchburchakli piramida asosining tomonlari 6, 8 va 10 ga teng. Piramidaning yon qirralari asosi tekisligi bilan bir xil burchak hosil qiladi. Agar piramidaning balandligi 4 ga teng bo'lsa, yon qirradi qanchaga teng bo'ladi?	$*\sqrt{41}$	3	4	7
144.	2	Silindr yon sirtining yuzi 24π ga, hajmi esa 48π ga teng. Silindrning balandligini toping.	*3	4	5	6